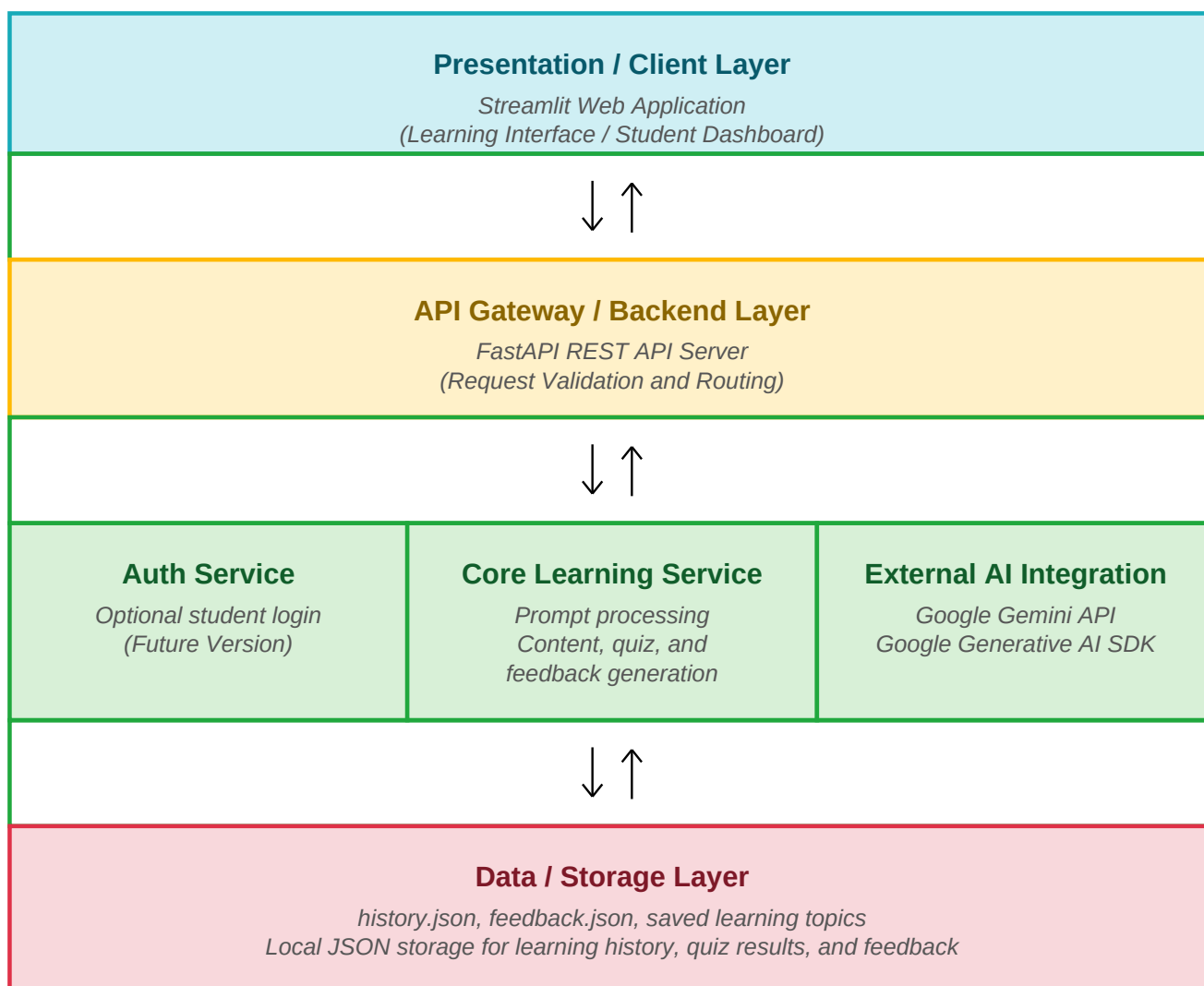


Solution Architecture

Date	30 June 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the student interface for entering learning questions, selecting topics, viewing AI explanations, taking quizzes, and tracking progress.	Streamlit, HTML, CSS
API Gateway	Handles incoming API requests, validates input, and routes learning requests to appropriate application services.	FastAPI, Uvicorn
Learning Services	Processes learner prompts, generates explanations and summaries, creates quizzes, evaluates answers, and logs feedback.	Python, Google Generative AI SDK, Pydantic
Database / Storage	Stores learning history, saved topics, quiz results, and user feedback for future access and analysis.	JSON Files (history.json, feedback.json)
External AI Integration	Generates personalized learning content and contextual assistance using a generative AI model.	Google Gemini API
Testing Layer	Validates API endpoints, learning services, and user-facing workflows.	PyTest